

Adult Income Prediction — Model Comparison Report

Feature Engineering, Cross-Validation & Model Selection Analysis

1. Engineered Features

Feature	Type	Creation Rule	MI Score
age_bucket	Categorical	Binned age into 5 groups	0.0628
hours_bucket	Categorical	Binned hours into 4 groups	0.0302
has_capital_gain	Binary	Flag if capital_gain > 0	0.0095
log_capital_gain	Numeric	log(capital_gain + 1)	0.0775
higher_education	Binary	education-num >= 13	0.0495
edu_hours_interaction	Numeric	education-num x hours-per-week	0.0810

Key Insight: log_capital_gain and edu_hours_interaction have the highest predictive signal among engineered features.

2. Cross-Validated Model Comparison (5-fold)

Model	Accuracy	F1	ROC AUC
Logistic Regression	0.8557 ± 0.0022	0.6683 ± 0.0067	0.9120 ± 0.0024
Random Forest	0.8515 ± 0.0011	0.6646 ± 0.0038	0.9020 ± 0.0028
Gradient Boosting	0.8638 ± 0.0032	0.6790 ± 0.0071	0.9200 ± 0.0024

Observations:

- Gradient Boosting performs best across all metrics.
- Logistic Regression has strong ROC AUC (0.9120) and is highly interpretable.
- All models are stable (low standard deviations across folds).

3. Statistical Comparison

Comparison	Test	p-value
Logistic vs Random Forest	Paired t-test	0.0002
Logistic vs Gradient Boosting	Paired t-test	0.0048
Random Forest vs Gradient Boosting	Paired t-test	0.0002

Conclusion: Gradient Boosting is statistically significantly better than both Logistic Regression and Random Forest ($p < 0.05$). The difference is meaningful and not due to random chance.

4. Feature Importance Analysis

Top 5 Features (Random Forest):

- num__fnlwgt (0.1428) — Demographic weight
- num__age (0.1088) — Age
- num__edu_hours_interaction (0.0711) — Engineered interaction
- cat__marital-status_Married-civ-spouse (0.0643) — Marital status
- num__capital-gain (0.0528) — Capital gain

Engineered Features that Mattered Most:

- log_capital_gain (0.0421) — Handles skewness of capital gain
- edu_hours_interaction (0.0711) — Captures combined effect of education and hours
- higher_education (0.0221) — Strong predictor of high income

5. Logistic Regression Coefficients

Top Positive Coefficients:

- log_capital_gain (6.01) — Strongest predictor
- marital-status_Married-civ-spouse (1.60)
- native-country_Ireland (1.21)
- capital_gain (1.16)
- education-num (0.84)

Top Negative Coefficients:

- age_bucket_<25 (-1.36) — Young age → lower income
- Priv-house-serv (-1.21)
- Never-married (-1.05)
- native-country_Trinidad&Tobago (-1.06)

Interpretation: Higher capital gain, being married, and higher education strongly predict high income. Young age and private house service predict lower income.

6. Feature Selection Impact

Model	ROC AUC (All)	ROC AUC (Selected, k=20)	Change
Random Forest	0.9020	0.9018	~ -0.02%

Decision for Day 4: Keep all features.

- Performance drop from feature selection is minimal (< 1%).
- Engineered features provide valuable information for hyperparameter tuning.
- More features give the model more patterns to learn.

7. Recommended Model for Hyperparameter Tuning

Primary Model: Gradient Boosting

- Best ROC AUC (0.9200) and F1 (0.6790)
- Handles non-linear relationships naturally
- Stable across folds (low std deviation)

Secondary Model: Logistic Regression

- Highly interpretable (coefficients)
- Fast to train
- Good baseline for comparison

8. Features to Keep for Day 4

Category	Features
Original Numeric	age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week
Original Categorical	workclass, education, marital-status, occupation, relationship, race, sex, native-country
Engineered	age_bucket, hours_bucket, has_capital_gain, log_capital_gain, higher_education, edu_hours_interaction

9. Next Steps (Day 4)

- Hyperparameter tuning using GridSearchCV
- Focus on Gradient Boosting — `n_estimators`: [50, 100, 200], `learning_rate`: [0.01, 0.05, 0.1], `max_depth`: [3, 5, 7]
- Compare tuned models using cross-validation
- Final evaluation on hold-out test set